Rangan Pratik Borah

B.Tech — Electronics & Communication Engineering | 2nd Semester | VIT, Vellore

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CAREER OBJECTIVE

Eager to contribute to real-world projects in electronics, embedded systems, web development or AI/software development while continuously learning and improving technical skills to gain practical industry exposure.

TECHNICAL SKILLS

Programming Languages	Python, Java
AI & Machine Learning	OpenCV, NumPy, face-recognition, Haar Cascade, LBPH, 128-D feature vectors, Euclidean & Cosine similarity
Embedded Systems	Arduino, ESP32, Raspberry Pi, sensor integration, IoT automation
Web Development	HTML (basics)
Databases	SQL (SQLite / MySQL) used for face-matching pipelines
Hardware & Design	Logic Design, Intel Quartus Prime, Half Adder circuit design, Electronics fundamentals
Tools & Platforms	GitHub, VS Code, Arduino IDE, Eclipse IDE, Spyder
Soft Skills	Problem-solving, Teamwork, Hackathon experience, Fast learner, Adaptable

PROJECTS

1. MouFace NEXUS AI – Face Recognition System (Duration: 4 Days)

GITHUB LINK: https://github.com/RanganBorah/MouFace-NEXUS-AI

ngrok web LINK: https://irately-curliest-dennis.ngrok-free.dev

Tech Stack: Java, Spring Boot, OpenCV, MySQL, Haar Cascade, LBPH, HTML, CSS, JavaScript

Description-

- Built a real-time face recognition system with web-based registration and live detection
- Used Haar Cascade for detection and LBPH for recognition with optimized accuracy
- Integrated MySQL for user data management and real-time retrieval
- Developed full pipeline: capture → training → recognition → display
- Deployed live demo access using **ngrok** for remote usage
- The web application is hosted via ngrok and maybe temporarily inactive. It can be brought online anytime as required. Please verify availability when accessing.

2. AI-Based Smart Irrigation System (Duration: 2 Days)

Tech Stack: Arduino, ESP32, Sensors

Description-

- Developed an IoT-based irrigation system using real-time soil and environmental data
- Integrated AI to take watering decisions, reducing unnecessary water usage
- Implemented sensor integration and control logic on ESP32 for end-to-end automation

3. Passcode-Based Electronic Security System (Duration: 1 Day)

Tech Stack: Arduino, ESP 32

Description-

- Built a passcode-protected security system with real-time input validation
- Implemented hardware-software integration with feedback via buzzer and LEDs

4. Half Adder Circuit | Digital Logic Design (Duration: 1 Day)

Tech Stack: Intel Quartus Prime

Description-

- Designed and simulated a half adder circuit demonstrating combinational logic fundamentals

EDUCATION

Bachelor of Technology — Electronics & Communication Engineering | 2025 – 2029 (Currently in 1st year)
Vellore Institute of Technology (VIT), Vellore | GPA: 8.6 / 10 (1st Semester)

- Relevant coursework: AI/ML, Python, Java, Logic Design, Management
- Active participant in hackathons and technical competitions

KEY ACHIEVEMENTS

- Achieved 90%+ accuracy in a real-time face recognition system
- Built **AI + Embedded** Systems projects in first year
- Hands-on experience in IoT, computer vision and hardware-software integration
- Active participant in hackathons and rapid prototyping environments

HACKATHONS & ACTIVITIES

Robotics Club – Member | Hands-on experience working with robotics hardware, embedded electronics and engineering projects

Hackathon Participant - Multiple Events

- Collaborated in team-based hackathons developing end-to-end solutions under time constraints
- Experienced rapid prototyping, technical ideation and presenting solutions to judges